

Letters and Principal Doctrines

ΕΠΙΣΤΟΛΑΙ ΚΑΙ ΚΥΡΙΑΙ ΔΟΞΑΙ

Epicurus

translated by C. D. Yonge (1853)

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Letter to Herodotus

On the principles of nature — an epitome of the physics

10.35

“For those, O Herodotus, who are not able accurately to comprehend all the things which I have written about nature, nor to investigate those larger books which I have composed on the subject, I have made an abridgment of the whole discussion on this question, as far as I thought sufficient to enable them to recollect accurately the most fundamental points; that so, on all grave occasions, they might be able to assist themselves on the most important and undeniable principles; in proportion as they devoted themselves to speculations on natural philosophy. And here it is necessary for those who have made sufficient progress in their view of the general question, to recollect the principles laid down as elements of the whole discussion; for we have still greater need of a correct notion of the whole, than we have even of an accurate understanding of the details. We must, therefore, give preference to the former knowledge, and lay up in our memory those principles on which we may rest, in order to arrive at an exact perception of things, and at a certain knowledge of particular objects.

10.36

“Now one has arrived at that point when one has thoroughly embraced the conceptions, and, if I may so express myself, the most essential forms, and when one has impressed them adequately on one’s senses. For this clear and precise knowledge of the whole, taken together, necessarily facilitates one’s particular perceptions, when one has brought one’s ideas back to the elements and simple terms. In short, a veritable synthesis, comprising the entire circle of the phænomena of the universe, ought to be able to resume in itself, and in a few words, all the particular facts which have been previously studied. This method being useful even to those who are already familiarised with the laws of the universe, I recommend them, while

still pursuing without intermission the study of nature, which contributes more than anything else to the tranquillity and happiness of life, to make a concise statement or summary of their opinions.

10.38

“First of all, then, Herodotus, one must determine with exactness the notion comprehended under each separate word, in order to be able to refer to it, as to a certain criterion, the conceptions which emanate from ourselves, the ulterior researches and the difficulties; otherwise the judgment has no foundation. One goes on from demonstration to demonstration ad infinitum; or else one gains nothing beyond mere words. In fact, it is absolutely necessary that in every word we should perceive directly, and without the assistance of any demonstration, the fundamental notion which it expresses, if we wish to have any foundation to which we may refer our researches, our difficulties, and our personal judgments, whatever in other respects may be the criterion which we adopt, whether we take as our standard the impressions produced on our senses, or the actual impression in general; or whether we cling to the idea by itself, or to any other criterion.

“We must also note carefully the impressions which we receive in the presence of objects, in order to bring ourselves back to that point in the circumstances in which it is necessary to suspend the judgment, or even when the question is about things, the evidence of which is not immediately perceived.

10.39

“When these foundations are once laid we may pass to the study of those things, the evidence of which is not immediate. And, first of all, we must admit that nothing can come of that which does not exist; for, were the fact otherwise, then every thing would be produced from everything, and there would be no need of any seed. And if that which disappeared were so absolutely destroyed as to become non-existent, then every thing would soon perish, as the things with which they would be dissolved would have no existence. But, in truth, the

universal whole always was such as it now is, and always will be such. For there is nothing into which it can change; for there is nothing beyond this universal whole which can penetrate into it, and produce any change in it.”

10.40

(And Epicurus establishes the same principles at the beginning of the great Abridgment; and in the first book of his treatise on Nature.) “Now the universal whole is a body; for our senses bear us witness in every case that bodies have a real existence; and the evidence of the senses, as I have said before, ought to be the rule of our reasonings about everything which is not directly perceived. Otherwise, if that which we call the vacuum, or space, or intangible nature, had not a real existence, there would be nothing on which the bodies could be contained, or across which they could move, as we see that they really do move. Let us add to this reflection that one cannot conceive, either in virtue of perception, or of any analogy founded on perception, any general quality peculiar to all beings which is not either an attribute, or an accident of the body, or of the vacuum.”

10.41

(The same principles are laid down in the first, and fourteenth, and fifteenth book of the treatise on Nature; and also in the Great Abridgment.)

“Now, of bodies, some are combinations, and some the elements out of which these combinations are formed. These last are indivisible, and protected from every kind of transformation; otherwise everything would be resolved into non-existence. They exist by their own force, in the midst of the dissolution of the combined bodies, being absolutely full, and as such offering no handle for destruction to take hold of. It follows, therefore, as a matter of absolute necessity, that the principles of things must be corporeal, indivisible elements.

10.42

“The universe is infinite. For that which is finite has an extreme, and that which has an extreme is looked at in relation to something else.

Consequently, that which has not an extreme, has no boundary; and if it has no boundary, it must be infinite, and not terminated by any limit. The universe then is infinite, both with reference to the quantity of bodies of which it is made up, and to the magnitude of the vacuum; for if the vacuum were infinite, the bodies being finite, then, the bodies would not be able to rest in any place; they would be transported about, scattered across the infinite vacuum for want of any power to steady themselves, or to keep one another in their places by mutual repulsion. If, on the other hand, the vacuum were finite, the bodies being infinite, then the bodies clearly could never be contained in the vacuum.

“Again: the atoms which form the bodies, these full elements from which the combined bodies come, and into which they resolve themselves, assume an incalculable variety of forms, for the numerous differences which the bodies present cannot possibly result from an aggregate of the same forms. Each variety of forms contains an infinity of atoms, but there is not for that reason an infinity of atoms; it is only the number of them which is beyond all calculation.”

10.43

(Epicurus adds, a little lower down, that divisibility, ad infinitum, is impossible; for, says he, the only things which change are the qualities; unless, indeed, one wishes to proceed from division to division, till one arrives absolutely at infinite littleness.)

“The atoms are in a continual state of motion.”

(He says, farther on, that they move with an equal rapidity from all eternity, since the vacuum offers no more resistance to the lightest than it does to the heaviest.)

10.44

“Among the atoms, some are separated by great distances, others come very near to one another in the formation of combined bodies, or at times are enveloped by others which are combining; but in this latter case they, nevertheless, preserve their own peculiar motion, thanks to

the nature of the vacuum, which separates the one from the other, and yet offers them no resistance. The solidity which they possess causes them, while knocking against one another, to re-act the one upon the other; till at last the repeated shocks bring on the dissolution of the combined body; and for all this there is no external cause, the atoms and the vacuum being the only causes.”

10.45

(He says, further on, that the atoms have no peculiar quality of their own, except from magnitude and weight. As to colour, he says in the twelfth book of his *Principia*, that it varies according to the position of the atoms. Moreover, he does not attribute to the atoms any kind of dimensions; and, accordingly, no atom has ever been perceived by the senses; but this expression, if people only recollect what is here said, will by itself offer to the thoughts a sufficient image of the nature of things.)

“But, again, the worlds also are infinite, whether they resemble this one of ours or whether they are different from it. For, as the atoms are, as to their number, infinite, as I have proved above, they necessarily move about at immense distances; for besides, this infinite multitude of atoms, of which the world is formed, or by which it is produced, could not be entirely absorbed by one single world, nor even by any worlds, the number of which was limited, whether we suppose them like this world of ours, or different from it. There is, therefore, no fact inconsistent with an infinity of worlds.

10.46

“Moreover, there are images resembling, as far as their form goes, the solid bodies which we see, but which differ materially from them in the thinness of their substance. In fact it is not impossible but that there may be in space some secretions of this kind, and an aptitude to form surfaces without depth, and of an extreme thinness; or else that from the solids there may emanate some particles which preserve the connection, the disposition, and the motion which they had in the body. I give the name of images to these representations; and, indeed,

their movement through the vacuum taking place, without meeting any obstacle or hindrance, perfects all imaginable extent in an inconceivable moment of time; for it is the meeting of obstacles, or the absence of obstacles, which produces the rapidity or the slowness of their motion. At all events, a body in motion does not find itself, at any moment imaginable, in two places at the same time; that is quite inconceivable. From whatever point of infinity it arrives at some appreciable moment, and whatever may be the spot in its course in which we perceive its motion, it has evidently quitted that spot at the moment of our thought; for this motion which, as we have admitted up to this point, encounters no obstacle to its rapidity, is wholly in the same condition as that the rapidity of which is diminished by the shock of some resistance.

“It is useful, also, to retain this principle, and to know that the images have an incomparable thinness; which fact indeed is in no respect contradicted by sensible appearances. From which it follows that their rapidity also is incomparable; for they find everywhere an easy passage, and besides, their infinite smallness causes them to experience no shock, or at all events to experience but a very slight one, while an infinite multitude of elements very soon encounter some resistance.

10.48

“One must not forget that the production of images is simultaneous with the thought; for from the surface of the bodies images of this kind are continually flowing off in an insensible manner indeed, because they are immediately replaced. They preserve for a long time the same disposition, and the same arrangement that the atoms do in the solid body, although, notwithstanding, their form may be sometimes altered. The direct production of images in space is equally instantaneous, because these images are only light substances destitute of depth.

10.49

“But there are other manners in which natures of this kind are produced; for there is nothing in all this which at all contradicts the

senses, if one only considers in what way the senses are exercised, and if one is inclined to explain the relation which is established between external objects and ourselves. Also, one must admit that something passes from external objects into us in order to produce in us sight and the knowledge of forms; for it is difficult to conceive that external objects can affect us through the medium of the air which is between us and them, or by means of rays, whatever emissions proceed from us to them, so as to give us an impression of their form and colour. This phenomenon, on the contrary, is perfectly explained, if we admit that certain images of the same colour, of the same shape, and of a proportionate magnitude pass from these objects to us, and so arrive at being seen and comprehended. These images are animated by an exceeding rapidity, and, as on the other side, the solid object forming a compact mass, and comprising a vast quantity of atoms, emits always the same quantity of particles, the vision is continued, and only produces in us one single perception which preserves always the same relation to the object. Every conception, every sensible perception which bears upon the form or the other attributes of these images, is only the same form of the solid perceived directly, either in virtue of a sort of actual and continued condensation of the image, or in consequence of the traces which it has left in us.

10.51

“Error and false judgments always depend upon the supposition that a preconceived idea will be confirmed, or at all events will not be overturned, by evidence. Then, when it is not confirmed, we form our judgment in virtue of a sort of initiation of the thoughts connected, it is true with the perception, and with a direct representation; but still connected also with a conception peculiar to ourselves, which is the parent of error. In fact the representations which intelligence reflects like a mirror, whether one perceives them in a dream, or by any other conceptions of the intellect, or of any other of the criteria, can never resemble the objects that one calls real and true, unless there were objects of this kind perceived directly. And, on the other side, error

could not be possible if we did not receive some other motion also, a sort of initiative of intelligence connected; it is true with direct representation, but going beyond that representative. These conceptions being connected with the direct perception which produces the representation, but going beyond it, in consequence of a motion peculiar to the individual thought, produces error when it is not confirmed by evidence, or when it is contradicted by evidence; but when it is confirmed, or when it is not contradicted by evidence, then it produces truth.

10.52

“We must carefully preserve these principles in order not to reject the authority of the faculties which perceive truth directly; and not, on the other hand, to allow what is false to be established with equal firmness, so as to throw everything into confusion.

10.53

“Moreover, hearing is produced by some sort of current proceeding from something that speaks, or sounds, or roars, or in any manner causes any sort of audible circumstance. And this current is diffused into small bodies resembling one another in their parts; which, preserving not only some kind of relation between one another, but even a sort of particular identity with the object from which they emanate, puts us, very frequently, into a communication of sentiments with this object, or at least causes us to become aware of the existence of some external circumstance. If these currents did not carry with them some sort of sympathy, then there would be no such perception. We must not therefore think that it is the air which receives a certain form, under the action of the voice or of some other sound. For it is utterly impossible that the voice should act in this manner on the air. But the percussion produced in us when we, by the utterance of a voice, cause a disengagement of certain particles, constitutes a current resembling a light whisper, and prepares an acoustic feeling for us.

“We must admit that the case of smelling is the same as that of hearing. There would be no sense of smell if there did not emanate from most

objects certain particles capable of producing an impression on the smell. One class being ill-suited to the organ, and consequently producing a disordered state of it, the other being suited to it, and causing it no distress.

10.54

“One must also allow, that the atoms possess no one of the qualities of sensible objects, except form, weight, magnitude and anything else is unavoidably inherent in form; in fact, every quality is changeable, but the atoms are necessarily unchangeable; for it is impossible but that in the dissolution of combined bodies, there must be something which continues solid and indestructible, of such a kind, that it will not change either into what does not exist, or out of what does not exist; but that it results either from a simple displacement of parts, which is the most usual case, or from the addition or subtraction of certain particles. It follows from that, that that which does not admit of any change in itself, is imperishable, participates in no respect in the nature of changeable things, and in a word, has its dimensions and forms immutably determined. And this is proved plainly enough, because even in the transformations which take place under our eyes, in consequence of the retrenchment of certain parts, we can still recognise the form of these constituent parts; while those qualities, which are not constituent parts, do not remain like the form, but perish in the dissolution of the combination. The attributes which we have indicated, suffice to explain all the differences of combined bodies; for we must inevitably leave something indestructible, lest everything should resolve itself into non-existence.

10.56

“However, one must not believe that every kind of magnitude exists in atoms, lest we find ourselves contradicted by phænomena. But we must admit that there are atoms of different magnitude, because, as that is the case, it is then more easy to explain the impressions and sensations; at all events, I repeat, it is not necessary for the purpose of explaining

the differences of the qualities, to attribute to atoms every kind of magnitude.

10.57

“We must not suppose either, that an atom can become visible to us; for, first of all, one does not see that that is the case, and besides, one cannot even conceive, how an atom is to become visible; besides, we must not believe, that in a finite body there are particles of every sort, infinite in number; consequently, one must not only reject the doctrine of infinite divisibility in parcels smaller and smaller, lest we should be reducing everything to nothing, and find ourselves forced to admit, that in a mass composed of a crowd of elements, existence can reduce itself to non-existence. But one cannot even suppose that a finite object can be susceptible of transformations *ad infinitum*, or even of transformation into smaller objects than itself; for when once one has said that there are in an object particles of every kind, infinite in number, there is absolutely no means whatever of imagining that this object can have only a finite magnitude; in fact, it is evident that these particles, infinite in number, have some kind of dimension or other, and whatever this dimension may be in other respects, the objects which are composed of it will have an infinite magnitude; in presenting forms which are determined, and limits which are perceived by the senses, one conceives, easily, without its being necessary to study this last question directly, that this would be the consequence of the contrary supposition, and that consequently, one must come to look at every object as infinite.

10.58

“One must also admit, that the most minute particle perceptible to the sense, is neither absolutely like the objects which are susceptible of transformation, nor absolutely different from them. It has some characteristics in common with the object which admit of transformation, but it also differs from them, inasmuch as it does not allow any distinct parts to be discerned in it. When then, in virtue of these common characteristics, and of this resemblance, we wish to

form an idea of the smallest particle perceptible by the senses, in taking the objects which change for our terms of comparison, it is necessary that we should seize on some characteristic common to these different objects. In this way, we examine them successively, from the first to the last, not by themselves, nor as composed of parts in juxtaposition, but only in their extent; in other words, we consider the magnitudes by themselves, and in an abstract manner, inasmuch as they measure, the greater a greater extent, and the smaller a smaller extent. This analogy applies to the atom, as far as we consider it as having the smallest dimensions possible. Evidently by its minuteness, it differs from all sensible objects, still this analogy is applicable to it; in a word, we establish by this comparison, that the atom really has some extent, but we exclude all considerable dimensions, for the sake of only investing it with the smallest proportions.

“We must also admit, in taking for our guide, the reasoning which discourses to us things which are invisible to the senses, that the most minute magnitudes, those which are not compound magnitudes, and which from the limit of sensible extent, are the first measure of the other magnitudes which are only called greater or less in their relation to the others. For these relations which they maintain with these particles, which are not subject to transformation, suffice to give them this characteristic of first measure. But they cannot, like atoms, combine themselves, and form compound bodies in virtue of any motion belonging to themselves.

10.60

“Moreover, we must not say (while speaking of the infinite), that such or such a point is the highest point of it, or the lowest. For height and lowness must not be predicated of the infinite. We know, in reality, that if, wishing to determine the infinite, we conceive a point above our head, this point, whatever it may be, will never appear to us to have the character in question: otherwise, that which would be situated above the point so conceived as the limit of the infinite, would

be at the same moment, and by virtue of its relation to the same point, both high and low; and this is impossible to imagine.

“It follows that thought can only conceive that one single movement of transference, from low to high, *ad infinitum*; and one single movement from high to low. From low to high, when even the object in motion, going from us to the places situated above our heads, meets ten thousand times with the feet of those who are above us; and from high to low, when in the same way it advances towards the heads of those who are below us. For these two movements, looked at by themselves and in their whole, are conceived as really opposed the one to the other, in their progress towards the infinite.

10.61

“Moreover, all the atoms are necessarily animated by the same rapidity, when they move across the vacuum, or when no obstacle thwarts them. For why should heavy atoms have a more rapid movement than those which are small and light, since in no quarter do they encounter any obstacle? Why, on the other hand, should the small atoms have a rapidity superior to that of the large ones, since both the one and the other find everywhere an easy passage, from the very moment that no obstacle intervenes to thwart their movements? Movement from low to high, horizontal movement to and fro, in virtue of the reciprocal percussion of the atoms, movement downwards, in virtue of their weight, will be all equal, for in whatever sense the atom moves, it must have a movement as rapid as the thought, till the moment when it is repelled, in virtue of some external cause, or of its own proper weight, by the shock of some object which resists it.

10.62

“Again, even in the compound bodies, one atom does not move more rapidly than another. In fact, if one only looks at the continued movement of an atom which takes place in an indivisible moment of time, the briefest possible, they all have a movement equally rapid. At the same time, an atom has not, in any moment perceptible to the intelligence, a continued movement in the same direction; but rather a

series of oscillating movements from which there results, in the last analysis, a continued movement perceptible to the senses. If then, one were to suppose, in virtue of a reasoning on things invisible, that, in the intervals of time accessible to thought, the atoms have a continued movement one would deceive one's self, for that which is conceived by the thought is true as well as that which is directly perceived.

10.63

“Let us now return to the study of the affections, and of the sensations; for this will be the best method of proving that the soul is a bodily substance composed of slight particles, diffused over all the members of the body, and presenting a great analogy to a sort of spirit, having an admixture of heat, resembling at one time one, and at another time the other of those two principles. There exists in it a special part, endowed with an extreme mobility, in consequence of the exceeding slowness of the elements which compose it, and also in reference to its more immediate sympathy with the rest of the body. That it is which the faculties of the soul sufficiently prove, and the passions, and the mobility of its nature, and the thoughts, and, in a word, everything, the privation of which is death. We must admit that it is in the soul most especially that the principle of sensation resides. At the same time, it would not possess this power if it were not enveloped by the rest of the body which communicates it to it, and in its turn receives it from it, but only in a certain measure; for there are certain affections of the soul of which it is not capable.

10.65

“It is on that account that, when the soul departs, the body is no longer possessed of sensation; for it has not this power, (that of sensation namely) in itself; but, on the other hand, this power can only manifest itself in the soul through the medium of the body. The soul, reflecting the manifestations which are accomplished in the substance which environs it, realises in itself, in a virtue or power which belongs to it, the sensible affections, and immediately communicates them to the body in virtue of the reciprocal bonds of sympathy which unite it

to the body; that is the reason why the destruction of a part of the body does not draw after it a cessation of all feeling in the soul while it resides in the body, provided that the senses still preserve some energy; although, nevertheless, the dissolution of the corporeal covering, or even of any one of its portions, may sometimes bring on with it the destruction of the soul.

10.66

“The rest of the body, on the other hand, even when it remains, either as a whole, or in any part, loses all feeling by the dispersion of that aggregate of atoms, whatever it may be, that forms the soul. When the entire combination of the body is dissolved, then the soul too is dissolved, and ceases to retain those faculties which were previously inherent in it, and especially the power of motion; so that sensation perishes equally as far as the soul is concerned; for it is impossible to imagine that it still feels, from the moment when it is no longer in the same conditions of existence, and no longer possesses the same movements of existence in reference to the same organic system; from the moment, in short, when the things which cover and surround it are no longer such, that it retains in them the same movements as before.

(Epicurus expresses the same ideas in other works, and adds that the soul is composed of atoms of the most perfect lightness and roundness; atoms wholly different from those of fire. He distinguishes in it the irrational part which is diffused over the whole body, from the rational part which has its seat in the chest, as is proved by the emotions of fear and joy. He adds that sleep is produced when the parts of the soul diffused over the whole of the body concentrate themselves, or when they disperse and escape by the pores of the body; for particles emanate from all bodies.)

10.67

“It must also be observed, that I use the word incorporeal (ἄσώματος) in the usual acceptation of the word, to express that which is in itself conceived as such. Now, nothing can be conceived in itself as

incorporeal except the vacuum; but the vacuum cannot be either passive or active; it is only the condition and the place of movement. Accordingly, they who pretend that the soul is incorporeal, utter words destitute of sense; for, if it had this character, it would not be able either to do or to suffer anything; but, as it is, we see plainly enough that it is liable to both these circumstances.

10.68

“Let us then apply all these reasonings to the affections and sensations, recollecting the ideas which we laid down at the beginning, and then we shall see clearly that these general principles contain an exact solution of all the particular cases.

10.69

“As to forms, and hues, and magnitudes, and weight, and the other qualities which one looks upon as attributes, whether it be of every body, or of those bodies only which are visible and perceived by the senses, this is the point of view under which they ought to be considered: they are not particular substances, having a peculiar existence of their own, for that cannot be conceived; nor can one say any more that they have no reality at all. They are not incorporeal substances inherent in the body, nor are they parts of the body. But they constitute by their union the eternal substance and the essence of the entire body. We must not fancy, however, that the body is composed of them, as an aggregate is formed of particles of the smallest dimensions of atoms or magnitudes, whatever they may be, smaller than the compound body itself; they only constitute by their union, I repeat, the eternal substance of the body. Each of these attributes has ideas and particular perceptions which correspond to it; but they cannot be perceived independently of the whole subject taken entirely; the union of all these perceptions forms the idea of the body. Bodies often possess other attributes which are not eternally inherent in them, but which, nevertheless, cannot be ranged among the incorporeal and invisible things. Accordingly, it is sufficient to express the general idea of the movement of transference to enable us to

conceive in a moment certain distinct qualities, and those combined beings, which, being taken in their totality, receive the name of bodies; and the necessary and eternal attributes without which the body cannot be conceived.

10.70

“There are certain conceptions corresponding to these attributes; but, nevertheless, they cannot be known abstractedly, and independently of some subjects; and further, inasmuch as they are not attributes necessarily inherent in the idea of a body, one can only conceive them in the moment in which they are visible; they are realities nevertheless; and one must not refuse them being an existence merely because they have neither the characteristic of the compound beings to which we give the name of bodies, nor that of the eternal attributes. We should be equally deceived if we were to suppose that they have a separate and independent existence; for that is true neither of them nor of the eternal attributes. They are, as one sees plainly, accidents of the body; accidents which do not of necessity make any part of its nature; which cannot be considered as independent substances, but still to each of which sensation gives the peculiar character under which it appears to us.

10.72

“Another important question is that of time. Here we cannot apply any more the method of examination to which we submit other objects, which we study with reference to a given subject; and which we refer to the preconceptions which exist in ourselves. We must seize, by analogy, and going round the whole circle of things comprised under this general denomination of time—we must seize, I say—that essential character which causes us to say that a time is long or short. It is not necessary for that purpose to seek for any new forms of expression as preferable to those which are in common use; we may content ourselves with those by which time is usually indicated. Nor need we, as certain philosophers do, affirm any particular attribute of time, for that would be to suppose that its essence is the same as that of this

attribute. It is sufficient too seek for the ingredients of which this particular nature which we call time is composed, and for the means by which it is measured. For this we have no need of demonstration; a simple exposition is sufficient. It is, in fact, evident, that we speak of time as composed of days and nights, and parts of days and nights; passiveness and impassibility, movement and repose, are equally comprised in time. In short, it is evident that in connection with these different states, we conceive a particular property to which we give the name of time.

(Epicurus lays down the same principles in the second book of his treatise on Nature, and in his great Abridgment.)

“It is from the infinite that the worlds are derived, and all the finite aggregates which present numerous analogies with the things which we observe under our own eyes. Each of these objects, great and small, has been separated from the infinite by a movement peculiar to itself. On the other hand, all these bodies will be successively destroyed, some more, and others less rapidly; some under the influence of one cause, and others because of the agency of some other.

10.74

(It is evident, after this, that Epicurus regards the worlds as perishable, since he admits that their parts are capable of transformation. He also says in other places, that the earth rests suspended in the air.)

“We must not believe that the worlds have of necessity all one identical form.

(He says, in fact, in the twelfth book of his treatise on the World, that the worlds differ from one another; some being spherical, other elliptical, and others of other shapes.)

“Nevertheless, there are not worlds of every possible form and shape.

“Let us also beware of thinking that animals are derived from the infinite; for there is no one who can prove that the germs from which animals are born, and plants, and all the other objects which we

contemplate, have been brought from the exterior in such a world, and that this same world would not have been able to produce them of itself. This remark applies particularly to the earth.

10.75

“Again, we must admit that in many and various respects, nature is both instructed and constrained by circumstances themselves; and that reason subsequently makes perfect and enriches with additional discoveries the things which it has borrowed from nature; in some cases rapidly, and in others more slowly. And in some cases according to periods and times greater than those which proceed from the infinite; in other cases according to those which are smaller. So, originally it was only in virtue of express agreements that one gave names to things. But men whose ideas and passions varied according to their respective nations, formed these names of their own accord, uttering divers sounds produced by each passion, or by each idea, following the differences of the situations and of the peoples. At a later period one established in each nation, in a uniform manner, particular terms intended to render the relations more easy, and language more concise. Educated men introduced the notion of things not discoverable by the senses, and appropriated words to them when they found themselves under the necessity of uttering their thoughts; after this, other men, guided in every point by reason, interpreted these words in the same sense.

10.77

“As to the heavenly phænomena, such as the motion and course of the stars, the eclipses, their rising and setting, and all other appearances of the same kind, we must beware of thinking that they are produced by any particular being which has regulated, or whose business it is to regulate, for the future, the order of the world, a being immortal and perfectly happy; for the cares and anxieties, the benevolence and the anger, far from being compatible with felicity, are, on the contrary, the consequence of weakness, of fear, and of the want which a thing has of something else. We must not fancy either that these globes of fire,

which roll on in space, enjoy a perfect happiness, and give themselves, with reflection and wisdom, the motions which they possess. But we must respect the established notions on this subject, provided, nevertheless, that they do not all contradict the respect due to truth; for nothing is more calculated to trouble the soul than this strife of contradictory notions and principles. We must therefore admit that from the first movement impressed on the heavenly bodies since the organization of the world there is derived a sort of necessity which regulates their course to this day.

10.78

“Let us be well assured that it is to physiology that it belongs to determine the causes of the most elevated phænomena, and that happiness consists, above all things, in the science of the heavenly things and their nature, and in the knowledge of analogous phænomena which may aid us in the comprehension of the ethics. These heavenly phænomena admit of several explanations; they have no reason of a necessary character, and one may explain them in different manners. In a word, they have no relation—a moment’s consideration will prove this by itself—with those imperishable and happy natures which admit of no division and of no confusion. As for the theoretical knowledge of the rising and setting of the stars, of the movement of the sun between the tropics, of the eclipses, and all other similar phænomena, that is utterly useless, as far as any influence upon happiness that it can have. Moreover, those who, though possessed of this knowledge, are ignorant of nature, and of the most probable causes of the phænomena, are no more protected from fear than if they were in the most complete ignorance; they even experience the most lively fears, for the trouble, with which the knowledge of which they are possessed inspires them, can find no issue, and is not dissipated by a clear perception of the reasons of these phænomena.

10.80

“As to us, we find many explanations of the motions of the sun, of the rising and setting of the stars, of the eclipses and similar phænomena,

just as well as of the more particular phænomena. And one must not think that this method of explanation is not sufficient to procure happiness and tranquillity. Let us content ourselves with examining how it is that similar phænomena are brought about under our own eyes, and let us apply these observations to the heavenly objects and to everything which is not known but indirectly. Let us despise those people who are unable to distinguish facts susceptible of different explanations from others which can only exist and be explained in one single way. Let us disdain those men who do not know, by means of the different images which result from distance, how to give an account of the different appearances of things; who, in a word, are ignorant what are the objects which can excite any trouble in us. If, then, we know that such a phænomenon can be brought about in the same manner as another given phænomenon of the same character which does not inspire us with any apprehension; and if, on the other hand, we know that it can take place in many different manners, we shall not be more troubled at the sight of it than if we knew the real cause of it.

10.81

“We must also recollect that that which principally contributes to trouble the spirit of men is the persuasion which they cherish that the stars are beings imperishable and perfectly happy, and that then one’s thoughts and actions are in contradiction to the will of these superior beings; they also, being deluded by these fables, apprehend an eternity of evils, they fear the insensibility of death, as that could affect them. What do I say? It is not even belief, but inconsiderateness and blindness which govern them in every thing, to such a degree that, not calculating these fears, they are just as much troubled as if they had really faith in these vain phantoms. And the real freedom from this kind of trouble consists in being emancipated from all these things, and in preserving the recollection of all the principles which we have established, especially of the most essential of them. Accordingly, it is well to pay a scrupulous attention to existing phænomena and to the sensations, to the general sensations for general things, and to the

particular sensations for particular things. In a word, we must take note of this, the immediate evidence with which each of these judicial faculties furnishes us; for, if we attend to these points, namely, whence confusion and fear arise, we shall divine the causes correctly, and we shall deliver ourselves from those feelings, tracing back the heavenly phænomena to their causes, and also all the others which present themselves at every step, and inspire the common people with extreme terror.

10.83

“This, Herodotus, is a kind of summary and abridgment of the whole question of natural philosophy. So that, if this reasoning be allowed to be valid, and be preserved carefully in the memory, the man who allows himself to be influenced by it, even though he may not descend to a profound study of its details, will have a great superiority of character over other men. He will personally discover a great number of truths which I have myself set forth in my entire work; and these truths being stored in his memory, will be a constant assistance to him. By means of these principles, those who have descended into the details, and have studied the question sufficiently, will be able, in bringing in all their particular knowledge to bear on the general subject, to run over without difficulty almost the entire circle of the natural philosophy; those, on the other hand, who are not yet arrived at perfection, and who have not been able to hear me lecture on these subjects, will be able in their minds to run over the main of the essential notions, and to derive assistance from them for the tranquillity and happiness of life.”

This then is his letter on physics.

Letter to Pythocles

On the heavenly bodies

10.84

“Cleon has brought me your letter, in which you continue to evince towards me an affection worthy of the friendship which I have for you. You devote all your care, you tell me, to engraving in your memory those ideas which contribute to the happiness of life; and you entreat me at the same time to send you a simple abridgment and abstract of my ideas on the heavenly phænomena, in order that you may without difficulty preserve the recollection of them. For, say you, what I have written on this subject in my other works is difficult to recollect, even with continual study.

10.85

“I willingly yield to your desire, and I have good hope, that in fulfilling what you ask, I shall be useful too to many others, especially to those who are as yet novices in the real knowledge of nature, and to those to whom the perplexities and the ordinary affairs of life leave but little leisure. Be careful then to seize on those precepts thoroughly, engrave them deeply in your memory, and meditate on them with the abridgment addressed to Herodotus, which I also send you.

“Know then, that it is with the knowledge of the heavenly phænomena, both with those which are spoken of in contact with one another, and of those which have a spontaneous existence, as with every other science; it has no other aim but that freedom from anxiety, and that calmness which is derived from a firm belief.

10.86

“It is not good to desire what is impossible, and to endeavour to enunciate a uniform theory about everything; accordingly, we ought not here to adopt the method, which we have followed in our researches into Ethics, or in the solution of problems of natural philosophy. We there said, for instance, that there are no other things, except bodies and the vacuum, that the atoms are the principles of things, and so of the rest. In a word, we gave a precise and simple explanation of every fact, conformable to appearances.

10.87

“We cannot act in the same way with respect to the heavenly phænomena: these productions may depend upon several different causes, and we may give many different explanations on this subject, equally agreeing with the impressions of the senses. Besides, it is not here a question about reasoning on new principles, and of laying down, à priori, rules for the interpretation of nature; the only guides for us to follow are the appearances themselves; for that which we have in view is not a set of systems and vain opinions, but much rather a life exempt from every kind of inquietude.

“The heavenly phænomena do not inspire those who give different explanations of them, conformable with appearances, instead of explaining them by hypothesis, with any alarm. But if, abandoning hypothesis, one at the same time renounces the attempt to explain them by means of analogies founded on appearances, then one is placing one’s self altogether at a distance from the science of nature, in order to fall into fables.

10.88

“It is possible that the heavenly phænomena may present some apparent characters which appear to assimilate them to those phænomena which we see taking place around ourselves, without there being any real analogy at the bottom. For the heavenly phænomena may depend for their production on many different causes; nevertheless, we must observe the appearances presented by each, and we must distinguish the different circumstances which attach to them, and which can be explained in different manners by means of analogous phænomena which arise under our eyes.

10.89

“The world is a collection of things embraced by the heaven, containing the stars, the earth, and all visible objects. This collection, separated from the infinite, is terminated by an extremity, which is either rare, or dense, or revolving, or in a state of repose, or of a round, or triangular, or of some shape or other in fact, for it may be of any shape, the dissolution of which must bring the destruction of

everything which they embrace. In fact, it can take place in every sort of way, since there is not one of those things which are seen which testifies against this world in which we cannot detect any extremity; and that such worlds are infinite in number is easily seen, and also that such a world can exist both in the world and in the μετακόσμιον, as we call the space between the worlds, being a huge space made up of plenum and vacuum, but not, as some philosophers pretend, an immensity of space absolutely empty. This production of a world may be explained thus: seeds suitably appropriated to such an end may emanate either from one or from several worlds, or from the space that separates them; they flow towards a particular point where they become collected together and organized; after that, other germs come to unite them together in such a way as to form a durable whole, a basis, a nucleus to which all successive additions unite themselves.

“One must not content one’s self in this question with saying, as one of the natural philosophers has done, that there is a re-union of the elements, or a violent motion in the vacuum under the influence of necessity, and that the body which is thus produced increases until it comes to crash against some other; for this doctrine is contrary to appearances.

“The sun, the moon, and the other stars, were originally formed separately, and were afterwards comprehended in the entire total of the world. All the other objects which our world comprises, for instance, the earth and the sea, were also formed spontaneously, and subsequently gained size by the addition and violent movement of light substances, composed of elements of fire and air, or even of these two principles at once. This explanation, moreover, is in accordance with the impressions of the senses.

10.91

“As to the magnitude of the sun and of the other stars, it is as far as we are concerned, such as it appears to us to be.”

(This same doctrine is reproduced, and occurs again in the eleventh book of his treatise on Nature; where he says, “If the distance has made it lose its size, à fortiori, it would take away its brilliancy; for colour has not, any more than size, the property of traversing distance without alteration.”)

“But, considered by itself, the sun may be a little greater or a little smaller than it appears; or it may be just such as it looks; for that is exactly the case with the fires of common occurrence among men, which are perceived by the senses at a distance. Besides, all the difficulties on this subject will be easily explained if one attends to the clear evidence of the perceptions, as I have shown in my books about Nature.

10.92

“The rising and setting of the sun, of the moon, and of the stars, may depend on the fact of their becoming lighted up, and extinguished alternately, and in the order which we behold. One may also give other reasons for this phænomenon, which are not contradicted by any sensible appearances; accordingly, one might explain them by the passage of the stars above and below the earth, for the impressions of the senses agree also with this supposition.

10.93

“As to their motion, one may make that depend on the circular movement of the entire heaven. One may also suppose that the stars move, while the heaven itself is immovable; for there is nothing to prevent the idea that originally, before the formation of the world, they may have received, by the appointment of fate, an impulse from east to west, and that now their movement continues in consequence of their heat, as the fire naturally proceeds onwards in order to seek the aliment which suits it.

“The intertropical movements of the sun and moon may depend, either on the obliquity impressed by fate on the heaven at certain determined epochs, or on the resistance of the air, or on the fact that

these ignited bodies stand in need of being nourished by a matter suitable to their nature, and that this matter fails them; or finally, they may depend on the fact of their having originally received an impulse which compels them to move as they do describing a sort of spiral figure. The sensible evidence does not in the least contradict these different suppositions, and all those of the same kind which one can form, having always a due regard to what is possible, and bringing back each phænomenon to its analogous appearances in sensible facts, without disquieting one's self about the miserable speculations of the astronomers.

10.94

“The evacuations and subsequent replenishings of the moon may depend either on a conversion of this body, or on the different forms which the air when in a fiery state can adopt, or perhaps to the interposition of another body, or lastly, to some one of the causes by which one gives account of the analogous phænomena which pass under our eyes. Provided, however, that one does not obstinately adopt an exclusive mode of explanation; and that, for want of knowing what is possible for a man to explain, and what is inaccessible to his intelligence; one does not throw one's self into interminable speculations.

10.95

“It may also be possibly the case that the moon has a light of her own, or that she reflects that of the sun. For we see around us many objects which are luminous of themselves, and many others which have only a borrowed light. In a word, one will not be arrested by any of the celestial phænomena, provided that one always recollects that there are many explanations possible; that one examines the principles and reasons which agree with this mode of explanation, and that one does not proceed in accounting for the facts which do not agree with this method, to suffer one's self to be foolishly carried away, and to propose a separate explanation for each phænomenon, sometimes in one way, and sometimes in another.

10.96

“The appearance of a face in the orb of the moon, may depend either on a displacement of its parts, or on the interposition of some obstacle, or on any other cause capable of accounting for such an appearance. For one must not neglect to apply this same method to all the heavenly phænomena; for, from the moment when one comes to any point of contradiction to the evidence of the senses, it will be impossible to possess perfect tranquillity and happiness.

“The eclipses of the sun and moon may depend either on the fact that these stars extinguish themselves, a phænomenon which we often see produced under our eyes, or on the fact of other bodies, the earth, the heaven, or something else of the same kind interposing, between them and us. Besides, we must compare the different modes of explanation appropriate to phænomena, and recollect that it is not impossible that many causes may at one and the same time concur in their production.

10.97

(He says the same thing in the twelfth book of his treatise on Nature; and adds that the eclipses of the sun arise from the fact that it penetrates into the shade of the moon, to quit it again presently; and the eclipses of the moon from the fact of its entering into the shade of the earth. We also find the same doctrine asserted by Diogenes, the Epicurean, in the first book of his Select Opinions.)

“The regular and periodical march of these phænomena has nothing in it that ought to surprise us, if we only attend to the analogous facts which take place under our eyes. Above all things let us beware of making the Deity interpose here, for that being we ought to suppose exempt from all occupation and perfectly happy; otherwise we shall be only giving vain explanations of the heavenly phænomena, as has happened already to a crowd of authors. Not being able to recognize what is really possible, they have fallen into vain theories, in supposing that for all phænomena there was but one single mode of production, and in rejecting all other explanations which are founded on probability; they have adopted the most unreasonable opinions, for

want of placing in the front the study of the heavenly phænomena, and of sensible facts, which ought to serve to explain the first.

10.98

“The differences in the length of nights and days may arise from the fact that the passage of the sun above the earth is more or less rapid; and more or less slow, according to the length of the regions which it has to pass through. Or, again, to the fact that certain regions are passed through more rapidly than others, as is seen to be the case by our own eyes, in those things to which we can compare the heavenly phænomena. As to those who on this point admit only one explanation as possible, they put themselves in opposition to facts, and lose sight of the bounds set to human knowledge.

10.99

“The prognostics which are derived from the stars may, like those which we borrow from animals, arise from a simple coincidence. They may also have other causes, for example, some change in the air; for these two suppositions both harmonize equally with facts; but it is impossible to distinguish in what case one is to attribute them to the one cause or to the other.

10.100

“The clouds may be formed either by the air condensed under the pressure of the winds, or by the agency of atoms set apart for that end, or by emanations from the earth and waters, or by other causes. For there are a great number which are all equally able to produce this effect. When the clouds clash with one another, or undergo any transformation, they produce showers; and the long rains are caused by the motion of the clouds when moved from places suitable to them through the air, when a more violent inundation than usual takes place, from collections of some masses calculated to produce these effects.

“Thunder possibly arises from the movement of the winds revolving in the cavities of the clouds; of which we may see an image in vessels in our own daily use. It may also arise from the noise of fire acted upon by

the wind in them, and from the tearings and ruptures of the clouds when they have received a sort of crystalline consistency. In a word, experience drawn from our senses, teaches us that all these phænomena, and that one in particular, may be produced in many different manners.

10.101

“One may also assign different causes to the lightning; either the shock and collision of the clouds produce a fiery appearance, which is followed by lightning; or the lighting up of the clouds by the winds, produces this luminous appearance; or the mutual pressure of the clouds, or that of the wind against them, disengages the lightning. Or, one might say, that the interception of the light diffused from the stars, arrested for a time in the bosom of the clouds, is driven from them subsequently by their own movements, and by those of the winds, and so escapes from their sides; that the lightning is an extremely subtile light that evaporates from the clouds; that the clouds which carry the thunder are collected masses of fire; that the lightning arises from the motion of the fire, or from the conflagration of the wind, in consequence of the rapidity and continuousness of its motion. One may also attribute the luminous appearance of lightning to the rupture of the clouds under the action of the winds, or to the fall of inflammable atoms. Lastly, one may easily find a number of other explanations, if one applies to sensible facts, in order to search out the analogies which they present to the heavenly phænomena.

10.103

“Lightning precedes thunder, either because it is produced at the same moment that the wind falls on the cloud, while the noise is only heard at the instant when the wind has penetrated into the bosom of the cloud; or, perhaps, the two phænomena being simultaneous, the lightning arrives among us more rapidly than the noise of the thunder-bolt, as is in fact remarked in other cases when we see at a distance the clash of two objects.

10.104

“The thunderbolt may be produced either by a violent condensation of the winds, or by their rapid motion and conflagration. It may arise from the fact of the winds meeting in places which are too dense, in consequence of the accumulation of clouds, and then a portion of the current detaches itself and proceeds towards the lower situations; or else it may be caused by the fire which is contained in the bosom of the clouds precipitating itself downwards. As one may suppose that an immense quantity of fire being accumulated in the clouds dilates, violently bursting the substance which envelops it, because the resistance of the centre hinders it from proceeding further. This effect is especially produced in the neighbourhood of high mountains; and, accordingly, they are very frequently struck with the thunderbolts. In short, one may give a number of explanations of the thunderbolt; but we ought, above all things, to be on our guard against fables, and this one will easily be, if one follows faithfully the sensible phænomena in the explanation of these things, which are not perceived, except indirectly.

10.105

“Hurricanes (πρηστῆρες) may be caused either by the presence of a cloud, which a violent wind sets in motion and precipitates with a spiral movement towards the lower regions, or by a violent gust which bears a cloud into the neighbourhood of some other current, or else by the mere agitation of the wind by itself, when air is brought together from the higher regions and compressed without being able to escape on either side, in consequence of the resistance of the air which surrounds it; when the hurricane descends towards the earth, then there result whirlwinds in proportion to the rapidity of the wind that has produced them; and this phænomenon extends over the sea also.

10.106

“Earthquakes may arise from the wind penetrating into the interior of the earth, or from the earth itself receiving incessantly the addition of exterior particles, and being in incessant motion as to its constituent atoms, being in consequence disposed to a general vibration. That

which permits the wind to penetrate is the fact that falls take place in the interior, or that the air being impressed by the winds insinuates itself into the subterraneous caverns. The movement which numberless falls and the re-action of the earth communicate to the earth, when this motion meets bodies of greater resistance and solidity, is sufficient to explain the earthquakes. One might, however, give an account of them in several other ways.

“Winds are caused, either by the successive and regular addition of some foreign matter, or else by the re-union of a great quantity of water; and the differences of the winds may arise from the fact that some portions of this same matter fall into the numerous cavities of the earth, and are divided there.

10.107

“Hail is produced by an energetic condensation acting on the ethereal particles which the cold embraces in every direction; or, in consequence of a less violent condensation acting however on aqueous particles, and accompanied by division, in such a manner as to produce, at the same time, the re-union of certain elements and of the collective masses; or by the rupture of some dense and compact mass which would explain at the same time, the numerousness of the particles and their individual hardness. As to the spherical form of the hail, one may easily account for that by admitting that the shocks which it receives in every direction make all the angles disappear, or else that at the moment when the different fragments are formed, each of them is equally embraced on all sides by aqueous or ethereal particles.

10.108

“Snow may be produced by a light vapour full of moisture which the clouds allow to escape by passages intended for that end, when they are pressed, in a corresponding manner, by other clouds, and set in motion by the wind. Subsequently, these vapours become condensed in their progress under the action of the cold which surrounds the clouds in the lower regions. It may also be the case that this phænomenon is produced by clouds of a slight density as they become condensed. In

this case the snow which escapes from the clouds would be the result of the contact, or approximation of the aqueous particles, which in a still more condensed state produce hail. This effect is most especially produced in the air. Snow, again, may result from the collection of clouds previously condensed and solidified; or from a whole army of other causes.

“Dew proceeds from a re-union of particles contained in the air calculated to produce this moist substance. These particles may be also brought from places which are moist or covered with water (for in those places, above all others, it is that dew is abundant). These then re-unite, again resume their aqueous form, and fall down. The same phænomenon takes place in other cases before our own eyes under many analogies.

10.109

“Hoar-frost is dew congealed by the influence of the cold air that surrounds it.

“Ice is formed either by the detrition of round atoms contained in the water, and the re-union at scalene and acute angles of the atoms which exist in the water, or by an addition from without of these latter particles, which penetrating into the water, solidify it by driving away an equal amount of round atoms.

10.110

“The rainbow may be produced by the reflection of the solar rays on the moist air; or it may arise from a particular property of light and air, in virtue of which these particular appearances of colour are formed, either because the shades which we perceive result directly from this property, or because, on the contrary, it only produces one single shade, which, reflecting itself on the nearest portions of the air, communicates to them the tints which we observe. As to the circular form of the rainbow, that depends either on the fact of the sight perceiving an equal distance in every direction, or the fact of the atoms taking this form when re-uniting in the air; or it may be caused

by its detaching from the air which moves towards the moon, certain atoms which, being re-united in the clouds, give rise to this circular appearance.

10.111

“The lunar halo arises from the fact of the air, which moves towards the moon from all quarters, uniformly intercepting the rays emitted by this star, in such a way as to form around it a sort of circular cloud which partially veils it. It may also arise from the fact of the moon uniformly rejecting from all quarters, the air which surrounds it, in such a manner as to produce this circular and opaque covering. And perhaps this opaqueness may be caused by some particles which some current brings from without; perhaps also, the heat communicates to the moon the property of emitting by the pores in its surface, the particles by which this effect is produced.

“Comets arise either from the fact, that in the circumstances already stated, there are partial conflagrations in certain points of the heaven; or, that at certain periods, the heaven has above our heads a particular movement which causes them to appear. It may also be the case, that being themselves endowed with a peculiar movement, they advance at the end of certain periods of time, and in consequence of particular circumstances, towards the places which we inhabit. The opposite reasons explain their disappearance.

10.112

“Certain stars return to the same point in accomplishing their revolutions; and this arises, not only as has been sometimes believed, from the fact of the pole of the world, around which they move, being immovable, but also from the fact that the gyrations of the air which surrounds them, hinder them from deviations like the wandering stars. Perhaps also, this may be caused by the fact, that except in the route in which they move, and in which we perceive them, they do not find any material suitable to their nature. One may also explain this phænomenon in many other manners, reasoning according to sensible facts; thus, it is possible that certain stars may be wandering because

that is the nature of their movements, and, for the same reason, others may be immoveable. It is also possible, that the same necessity which has originally given them their circular movement, may have compelled some to follow their orbit regularly, and have subjected others to an irregular progress; we may also suppose that the uniform character of the centre which certain stars traverse favour their regular march, and their return to a certain; and that in the case of others, on the contrary, the differences of the centre produce the changes which we observe. Besides, to assign one single cause to all these phænomena, when the experience of our senses suggests us several, is folly. It is the conduct of ignorant astronomers covetous of a vain knowledge, who, assigning imaginary causes to facts, wish to leave wholly to the Deity the care of the government of the universe.

10.114

“Some stars appear to be left behind by others in their progress; this arises either from the fact of their having a slower motion, though traversing the same circle; or, because, though they are drawn on by the same propelling power, they have, nevertheless, a movement proper to themselves in a contrary direction; or it may be caused by the fact that, though all are placed in the same sphere of movement, still some have more space to traverse, and others less. To give one uniform and positive explanation of all these facts, is not consistent with the conduct of any people but those who love to flash prodigies in the eyes of the multitude.

10.115

“Falling stars may be particles detached from the stars, or fragments resulting from their collision; they may also be produced by the fall of substances which are set on fire by the action of the wind; by the re-union of inflammable atoms which are made to come together so as to produce this effect by a sort of reciprocal attraction; or else by the movement which is produced in consequence of the re-union of atoms in the very place where they meet. It may also happen that the light vapours re-unite and become condensed under the form of clouds, that

they then take fire in consequence of their rotatory motion, and that, bursting the obstacles which surround them, they proceed towards the places whither the force by which they are animated drags them. In short, this phænomenon also may admit of a great number of explanations.

10.116

“The presages which are drawn from certain animals arise from a fortuitous concourse of circumstances; for there is no necessary connection between certain animals and winter. They do not produce it; nor is there any divine nature sitting aloft watching the exits of these animals, and then accomplishing signs of this kind. Nor can such folly as this occur to any being who is even moderately comfortable, much less to one which is possessed of perfect happiness.

“Imprint all these precepts in your memory, O Pythocles, and so you will easily escape fables, and it will be easy for you to discover other truths analogous to these. Above all, apply yourself to the study of general principles, of the infinite, and of questions of this kind, and to the investigation of the different criteria and of the passions, and to the study of the chief good, with a view to which we prosecute all our researches. When these questions are once resolved, all particular difficulties will be made plain to you. As to those who will not apply themselves to these principles, they will neither be able to give a good explanation of these same questions, nor to reach that end to which all our researches tend.”

XXVI. Such are his sentiments on the heavenly phænomena. But concerning the rules of life, and how we ought to choose some things, and avoid others, he writes thus. But first of all, let us go through the opinions which he held, and his disciples held about the wise man.

He said that injuries existed among men, either in consequence of hatred, or of envy, or of contempt, all which the wise man overcomes by reason. Also, that a man who has once been wise can never receive the contrary disposition, nor can he of his own accord invent such a

state of things as that he should be subjected to the dominion of the passions; nor can he hinder himself in his progress towards wisdom. That the wise man, however, cannot exist in every state of body, nor in every nation. That even if the wise man were to be put to the torture, he would still be happy. That the wise man will only feel gratitude to his friends, but to them equally whether they are present or absent. Nor will he groan and howl when he is put to the torture. Nor will he marry a wife whom the laws forbid, as Diogenes says, in his epitome of the Ethical Maxims of Epicurus. He will punish his servants, but also pity them, and show indulgence to any that are virtuous. They do not think that the wise man will ever be in love, nor that he will be anxious about his burial, nor that love is a passion inspired by the Gods, as Diogenes says in his twelfth book. They also assert that he will be indifferent to the study of oratory. Marriage, say they, is never any good to a man, and we must be quite content if it does no harm; and the wise man will never marry or beget children, as Epicurus himself lays it down, in his Doubts and in his treatises on Nature. Still, under certain circumstances of life, he will forsake these rules and marry. Nor will he ever indulge in drunkenness, says Epicurus, in his Banquet, nor will he entangle himself in affairs of state (as he says in his first book on Lives). Nor will he become a tyrant. Nor will he become a Cynic (as he says in his second book about Lives). Nor a beggar. And even, though he should lose his eyes, he will still partake of life (as he says in the same book).

The wise man will be subject to grief, as Diogenes says, in the fifth book of his Select Opinions; he will also not object to go to law. He will leave books and memorials of himself behind him, but he will not be fond of frequenting assemblies. He will take care of his property, and provide for the future. He will like being in the country, he will resist fortune, and will grieve none of his friends. He will show a regard for a fair reputation to such an extent as to avoid being despised; and he will find more pleasure than other men in speculations.

All faults are not equal. Health is good for some people, but a matter of indifference to others. Courage is a quality which does not exist by nature, but which is engendered by a consideration of what is suitable. Friendship is caused by one's wants; but it must be begun on our side. For we sow the earth; and friendship arises from a community of, and participation in, pleasures. Happiness must be understood in two senses; the highest happiness, such as is that of God, which admits of no increase; and another kind, which admits of the addition or abstraction of pleasures. The wise man may raise statues if it suits his inclination, if it does not it does not signify. The wise man is the only person who can converse correctly about music and poetry; and he can realise poems, but not become a poet.

It is possible for one wise man to be wiser than another. The wise man will also, if he is in need, earn money, but only by his wisdom; he will propitiate an absolute ruler when occasion requires, and will humour him for the sake of correcting his habits; he will have a school, but not on such a system as to draw a crowd about him; he will also recite in a multitude, but that will be against his inclination; he will pronounce dogmas, and will express no doubts; he will be the same man asleep and awake; and he will be willing even to die for a friend.

These are the Epicurean doctrines.

XXVII. We must now proceed to his letter:—

Letter to Menoeceus

On the happy life

10.122

“Let no one delay to study philosophy while he is young, and when he is old let him not become weary of the study; for no man can ever find the time unsuitable or too late to study the health of his soul. And he who asserts either that it is not yet time to philosophize, or that the

hour is passed, is like a man who should say that the time is not yet come to be happy, or that it is too late. So that both young and old should study philosophy, the one in order that, when he is old, he may be young in good things through the pleasing recollection of the past, and the other in order that he may be at the same time both young and old, in consequence of his absence of fear for the future.

10.123

“It is right then for a man to consider the things which produce happiness, since, if happiness is present, we have everything, and when it is absent, we do everything with a view to possess it. Now, what I have constantly recommended to you, these things I would have you do and practise, considering them to be the elements of living well. First of all, believe that God is a being incorruptible and happy, as the common opinion of the world about God dictates; and attach to your idea of him nothing which is inconsistent with incorruptibility or with happiness; and think that he is invested with everything which is able to preserve to him this happiness, in conjunction with incorruptibility. For there are Gods; for our knowledge of them is indistinct. But they are not of the character which people in general attribute to them; for they do not pay a respect to them which accords with the ideas that they entertain of them. And that man is not impious who discards the Gods believed in by the many, but he who applies to the Gods the opinions entertained of them by the many. For the assertions of the many about the Gods are not anticipations (προλήψεις), but false opinions (ὑπολήψεις). And in consequence of these, the greatest evils which befall wicked men, and the benefits which are conferred on the good, are all attributed to the Gods; for they connect all their ideas of them with a comparison of human virtues, and everything which is different from human qualities, they regard as incompatible with the divine nature.

10.125

“Accustom yourself also to think death a matter with which we are not at all concerned, since all good and all evil is in sensation, and since

death is only the privation of sensation. On which account, the correct knowledge of the fact that death is no concern of ours, makes the mortality of life pleasant to us, inasmuch as it sets forth no illimitable time, but relieves us for the longing for immortality. For there is nothing terrible in living to a man who rightly comprehends that there is nothing terrible in ceasing to live; so that he was a silly man who said that he feared death, not because it would grieve him when it was present, but because it did grieve him while it was future. For it is very absurd that that which does not distress a man when it is present, should afflict him when only expected. Therefore, the most formidable of all evils, death, is nothing to us, since, when we exist, death is not present to us; and when death is present, then we have no existence. It is no concern then either of the living or of the dead; since to the one it has no existence, and the other class has no existence itself. But people in general, at times flee from death as the greatest of evils, and at times wish for it as a rest from the evils in life. Nor is the not living a thing feared, since living is not connected with it: nor does the wise man think not living an evil; but, just as he chooses food, not preferring that which is most abundant, but that which is nicest; so too, he enjoys time, not measuring it as to whether it is of the greatest length, but as to whether it is most agreeable. And he who enjoins a young man to live well, and an old man to die well, is a simpleton, not only because of the constantly delightful nature of life, but also because the care to live well is identical with the care to die well. And he was still more wrong who said:—

10.127

“For if this really was his opinion why did he not quit life? for it was easily in his power to do so, if it really was his belief. But if he was joking, then he was talking foolishly in a case where it ought not to be allowed; and, we must recollect, that the future is not our own, nor, on the other hand, is it wholly not our own, I mean so that we can never altogether await it with a feeling of certainty that it will be, nor altogether despair of it as what will never be. And we must consider

that some of the passions are natural, and some empty; and of the natural ones some are necessary, and some merely natural. And of the necessary ones some are necessary to happiness, and others, with regard to the exemption of the body, from trouble; and others with respect to living itself; for a correct theory, with regard to these things, can refer all choice and avoidance to the health of the body and the freedom from disquietude of the soul. Since this is the end of living happily; for it is for the sake of this that we do everything, wishing to avoid grief and fear; and when once this is the case, with respect to us, then the storm of the soul is, as I may say, put an end to; since the animal is unable to go as if to something deficient, and to seek something different from that by which the good of the soul and body will be perfected.

10.129

“For then we have need of pleasure when we grieve, because pleasure is not present; but when we do not grieve, then we have no need of pleasure; and on this account, we affirm, that pleasure is the beginning and end of living happily; for we have recognized this as the first good, being connate with us; and with reference to it, it is that we begin every choice and avoidance; and to this we come as if we judged of all good by passion as the standard; and, since this is the first good and connate with us, on this account we do not choose every pleasure, but at times we pass over many pleasures when any difficulty is likely to ensue from them; and we think many pains better than pleasures, when a greater pleasure follows them, if we endure the pain for a time.

10.130

“Every pleasure is therefore a good on account of its own nature, but it does not follow that every pleasure is worthy of being chosen; just as every pain is an evil, and yet every pain must not be avoided. But it is right to estimate all these things by the measurement and view of what is suitable and unsuitable; for at times we may feel the good as an evil, and at times, on the contrary, we may feel the evil as good. And, we think, contentment a great good, not in order that we may never

have but a little, but in order that, if we have not much, we may make use of a little, being genuinely persuaded that those men enjoy luxury most completely who are the best able to do without it; and that everything which is natural is easily provided, and what is useless is not easily procured. And simple flavours give as much pleasure as costly fare, when everything that can give pain, and every feeling of want, is removed; and corn and water give the most extreme pleasure when any one in need eats them. To accustom one's self, therefore, to simple and inexpensive habits is a great ingredient in the perfecting of health, and makes a man free from hesitation with respect to the necessary uses of life. And when we, on certain occasions, fall in with more sumptuous fare, it makes us in a better disposition towards it, and renders us fearless with respect to fortune. When, therefore, we say that pleasure is a chief good, we are not speaking of the pleasures of the debauched man, or those which lie in sensual enjoyment, as some think who are ignorant, and who do not entertain our opinions, or else interpret them perversely; but we mean the freedom of the body from pain, and of the soul from confusion. For it is not continued drinkings and revels, or the enjoyment of female society, or feasts of fish and other such things, as a costly table supplies, that make life pleasant, but sober contemplation, which examines into the reasons for all choice and avoidance, and which puts to flight the vain opinions from which the greater part of the confusion arises which troubles the soul.

10.133

“Now, the beginning and the greatest good of all these things is prudence, on which account prudence is something more valuable than even philosophy, inasmuch as all the other virtues spring from it, teaching us that it is not possible to live pleasantly unless one also lives prudently, and honourably, and justly; and that one cannot live prudently, and honestly, and justly, without living pleasantly; for the virtues are connate with living agreeably, and living agreeably is inseparable from the virtues. Since, who can you think better than

that man who has holy opinions respecting the Gods, and who is utterly fearless with respect to death, and who has properly contemplated the end of nature, and who comprehends that the chief good is easily perfected and easily provided; and the greatest evil lasts but a short period, and causes but brief pain. And who has no belief in necessity, which is set up by some as the mistress of all things, but he refers some things to fortune, some to ourselves, because necessity is an irresponsible power, and because he sees that fortune is unstable, while our own will is free; and this freedom constitutes, in our case, a responsibility which makes us encounter blame and praise. Since it would be better to follow the fables about the Gods than to be a slave to the fate of the natural philosopher; for the fables which are told give us a sketch, as if we could avert the wrath of God by paying him honour; but the other presents us with necessity who is inexorable.

10.135

“And he, not thinking fortune a goddess, as the generality esteem her (for nothing is done at random by a God), nor a cause which no man can rely on, for he thinks that good or evil is not given by her to men so as to make them live happily, but that the principles of great goods or great evils are supplied by her; thinking it better to be unfortunate in accordance with reason, than to be fortunate irrationally; for that those actions which are judged to be the best, are rightly done in consequence of reason.

“Do you then study these precepts, and those which are akin to them, by all means day and night, pondering on them by yourself, and discussing them with any one like yourself, and then you will never be disturbed by either sleeping or waking fancies, but you will live like a God among men; for a man living amid immortal Gods, is in no respect like a mortal being.”

Principal Doctrines

The forty-three maxims (Kyriai Doxai)

M1

1. “That which is happy and imperishable, neither has trouble itself, nor does it cause it to anything; so that it is not subject to the feelings of either anger or gratitude; for these feelings only exist in what is weak.

(In other passages he says that the Gods are speculated on by reason, some existing according to number, and others according to some similarity of form, arising from the continual flowing on of similar images, perfected for this very purpose in human form.)

M2

2. “Death is nothing to us; for that which is dissolved is devoid of sensation, and that which is devoid of sensation is nothing to us.

M3

3. “The limit of the greatness of the pleasures is the removal of everything which can give pain. And where pleasure is, as long as it lasts, that which gives pain, or that which feels pain, or both of them, are absent.

M4

4. “Pain does not abide continuously in the flesh, but in its extremity it is present only a very short time. That pain which only just exceeds the pleasure in the flesh, does not last many days. But long diseases have in them more that is pleasant than painful to the flesh.

M5

5. “It is not possible to live pleasantly without living prudently, and honourably, and justly; nor to live prudently, and honourably, and justly, without living pleasantly. But he to whom it does not happen to live prudently, honourably, and justly, cannot possibly live pleasantly.

M6

6. “For the sake of feeling confidence and security with regard to men, and not with reference to the nature of government and kingly power being a good, some men have wished to be eminent and powerful, in order that others might attain this feeling by their means; thinking that so they would secure safety as far as men are concerned. So that, if the life of such men is safe, they have attained to the nature of good; but if it is not safe, then they have failed in obtaining that for the sake of which they originally desired power according to the order of nature.

M7

7. “No pleasure is intrinsically bad: but the efficient causes of some pleasures bring with them a great many perturbations of pleasure.

M8

8. “If every pleasure were condensed, if one may so say, and if each lasted long, and affected the whole body, or the essential parts of it, then there would be no difference between one pleasure and another.

M9

9. “If those things which make the pleasures of debauched men, put an end to the fears of the mind, and to those which arise about the heavenly bodies, and death, and pain; and if they taught us what ought to be the limit of our desires, we should have no pretence for blaming those who wholly devote themselves to pleasure, and who never feel any pain or grief (which is the chief evil) from any quarter.

M10

10. “If apprehensions relating to the heavenly bodies did not disturb us, and if the terrors of death have no concern with us, and if we had the courage to contemplate the boundaries of pain and of the desires, we should have no need of physiological studies.

M11

11. “It would not be possible for a person to banish all fear about those things which are called most essential, unless he knew what is the nature of the universe, or if he had any idea that the fables told about it

could be true; and therefore, it is, that a person cannot enjoy unmixed pleasure without physiological knowledge.

M12

12. "It would be no good for a man to secure himself safety as far as men are concerned, while in a state of apprehension as to all the heavenly bodies, and those under the earth, and in short, all those in the infinite.

M13

13. "Irresistible power and great wealth may, up to a certain point, give us security as far as men are concerned; but the security of men in general depends upon the tranquillity of their souls, and their freedom from ambition.

M14

14. "The riches of nature are defined and easily procurable; but vain desires are insatiable.

M15

15. "The wise man is but little favoured by fortune; but his reason procures him the greatest and most valuable goods, and these he does enjoy, and will enjoy the whole of his life.

M16

16. "The just man is the freest of all men from disquietude; but the unjust man is a perpetual prey to it.

M17

17. "Pleasure in the flesh is not increased, when once the pain arising from want is removed; it is only diversified.

M18

18. "The most perfect happiness of the soul depends on these reflections, and on opinions of a similar character on all those questions which cause the greatest alarm to the mind.

M19

19. "Infinite and finite time both have equal pleasure, if any one measures its limits by reason.

M20

20. "If the flesh could experience boundless pleasure, it would want to dispose of eternity.

M21

21. "But reason, enabling us to conceive the end and dissolution of the body, and liberating us from the fears relative to eternity, procures for us all the happiness of which life is capable, so completely that we have no further occasion to include eternity in our desires. In this disposition of mind, man is happy even when his troubles engage him to quit life; and to die thus, is for him only to interrupt a life of happiness.

M22

22. "He who is acquainted with the limits of life knows, that that which removes the pain which arises from want, and which makes the whole of life perfect, is easily procurable; so that he has no need of those things which can only be attained with trouble.

M23

23. "But as to the subsisting end, we ought to consider it with all the clearness and evidence which we refer to whatever we think and believe; otherwise, all things will be full of confusion and uncertainty of judgment.

M24

24. "If you resist all the senses, you will not even have anything left to which you can refer, or by which you may be able to judge of the falsehood of the senses which you condemn.

M25

25. "If you simply discard one sense, and do not distinguish between the different elements of the judgment, so as to know on the one hand, the induction which goes beyond the actual sensation, or, on the other, the actual and immediate notion; the affections, and all the conceptions of the mind which lean directly on the sensible representation, you will be imputing trouble into the other sense, and destroying in that quarter every species of criterion.

M26

26. “If you allow equal authority to the ideas, which, being only inductive, require to be verified, and to those which bear about them an immediate certainty, you will not escape error; for you will be confounding doubtful opinions with those which are not doubtful, and true judgments with those of a different character.

M27

27. “If, on every occasion, we do not refer every one of our actions to the chief end of nature, if we turn aside from that to seek or avoid some other object, there will be a want of agreement between our words and our actions.

M28

28. “Of all the things which wisdom provides for the happiness of the whole life, by far the most important is the acquisition of friendship.

M29

29. “The same opinion encourages man to trust that no evil will be everlasting, or even of long duration; as it sees that, in the space of life allotted to us, the protection of friendship is most sure and trustworthy.

M30

30. “Of the desires, some are natural and necessary, some natural, but not necessary, and some are neither natural nor necessary, but owe their existence to vain opinions.”

(Epicurus thinks that those are natural and necessary which put an end to pains, as drink when one is thirsty; and that those are natural but not necessary which only diversify pleasure, but do not remove pain, such as expensive food; and that these are neither natural nor necessary, which are such as crowns, or the erection of statues.)

M31

31. “Those desires which do not lead to pain, if they are not satisfied, are not necessary. It is easy to impose silence on them when they appear difficult to gratify, or likely to produce injury.

M32

32. “When the natural desires, the failing to satisfy which is, nevertheless, not painful, are violent and obstinate, it is a proof that there is an admixture of vain opinion in them; for then energy does not arise from their own nature, but from the vain opinions of men.

M33

33. “Natural justice is a covenant of what is suitable, leading men to avoid injuring one another, and being injured.

M34

34. “Those animals which are unable to enter into an argument of this nature, or to guard against doing or sustaining mutual injury, have no such thing as justice or injustice. And the case is the same with those nations, the members of which are either unwilling or unable to enter into a covenant to respect their mutual interests.

M35

35. “Justice has no independent existence; it results from mutual contracts, and establishes itself wherever there is a mutual engagement to guard against doing or sustaining mutual injury.

M36

36. “Injustice is not intrinsically bad; it has this character only because there is joined with it a fear of not escaping those who are appointed to punish actions marked with that character.

M37

37. “It is not possible for a man who secretly does anything in contravention of the agreement which men have made with one another, to guard against doing, or sustaining mutual injury, to believe that he shall always escape notice, even if he have escaped notice already ten thousand times; for, till his death, it is uncertain whether he will not be detected.

M38

38. “In a general point of view, justice is the same thing to every one; for there is something advantageous in mutual society. Nevertheless,

the difference of place, and divers other circumstances, make justice vary.

M39

39. "From the moment that a thing declared just by the law is generally recognized as useful for the mutual relations of men, it becomes really just, whether it is universally regarded as such or not.

M40

40. "But if, on the contrary, a thing established by law is not really useful for the social relations, then it is not just; and if that which was just, inasmuch as it was useful, loses this character, after having been for some time considered so, it is not less true that, during that time, it was really just, at least for those who do not perplex themselves about vain words, but who prefer, in every case, examining and judging for themselves.

M41

41. "When, without any fresh circumstances arising, a thing which has been declared just in practice does not agree with the impressions of reason, that is a proof that the thing was not really just. In the same way, when in consequence of new circumstances, a thing which has been pronounced just does not any longer appear to agree with utility, the thing which was just, inasmuch as it was useful to the social relations and intercourse of mankind, ceases to be just the moment when it ceases to be useful.

M42

42. "He who desires to live tranquilly without having any thing to fear from other men, ought to make himself friends; those whom he cannot make friends of, he should, at least, avoid rendering enemies; and if that is not in his power, he should, as far as possible, avoid all intercourse with them, and keep them aloof, as far as it is for his interest to do so.

M43

43. "The happiest men are they who have arrived at the point of having nothing to fear from those who surround them. Such men live

with one another most agreeably, having the firmest grounds of confidence in one another, enjoying the advantages of friendship in all their fulness, and not lamenting, as a pitiable circumstance, the premature death of their friends.”